

Introduction to Logic

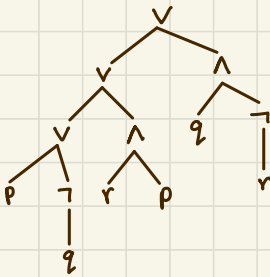
Assignment 3

Problem 1

For each formula below, rewrite the formula by inserting all the omitted parentheses and then draw a parse tree for it.

(a) $p \vee \neg q \vee r \wedge p \vee q \wedge \neg r$

$((p \vee (\neg q)) \vee (r \wedge p)) \vee (q \wedge (\neg r))$



Given the expression: $p \vee \neg q \vee r \wedge p \vee q \wedge \neg r$

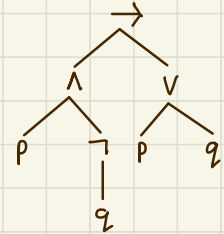
To add parentheses, we apply the precedence rules:

$p \vee (\neg q) \vee (r \wedge p) \vee (q \wedge \neg r)$

p	q	r	$\neg q$	$p \wedge \neg q$	$r \wedge p$	$p \vee (\neg q) \vee (r \wedge p)$	$((p \vee (\neg q)) \vee (r \wedge p)) \vee (q \wedge \neg r)$
T	T	T	F	F	T	T	T
T	T	F	F	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	F	T	T
F	T	T	F	F	T	T	T
F	T	F	F	F	F	F	F
F	F	T	T	F	T	T	T
F	F	F	T	F	F	F	T

(b) $(p \wedge (\neg q)) \rightarrow (p \vee q)$

$(p \wedge (\neg q)) \rightarrow (p \vee q)$



Problem 2

Suppose $\phi = ((p \wedge q) \rightarrow r) \leftrightarrow ((p \rightarrow r) \wedge (q \rightarrow r))$

(a) Describe a truth assignment which makes ϕ true.

(b) Describe a truth assignment which makes ϕ false.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

(a) $\phi = ((p \wedge q) \rightarrow r) \leftrightarrow ((p \rightarrow r) \wedge (q \rightarrow r))$

ϕ is true when all p, q and r are true.

(b) $\phi = ((p \wedge q) \rightarrow r) \leftrightarrow ((p \rightarrow r) \wedge (q \rightarrow r))$

ϕ is false when p is true and both q and r are false.

Problem 3

Determine by means of a truth table whether each of the following formula is satisfiable or valid or not.

(a) $p \rightarrow (q \rightarrow p)$

(b) $(p \rightarrow q) \rightarrow p$

(c) $(\top \rightarrow p) \wedge (p \rightarrow \perp)$

(a) $p \rightarrow (q \rightarrow p)$

p	q	$q \rightarrow p$	$p \rightarrow (q \rightarrow p)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	T

The given formula is both satisfiable and valid.

(b) $(p \rightarrow q) \rightarrow p$

p	q	$p \rightarrow q$	$(p \rightarrow q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	F

The given formula is satisfiable but not valid.

(c) $(\top \rightarrow p) \wedge (p \rightarrow \perp)$

T	p	$\perp$	$\top \rightarrow p$	$p \rightarrow \perp$	$(\top \rightarrow p) \wedge (p \rightarrow \perp)$
T	T	F	T	F	F
T	F	F	F	T	F

The given formula is neither satisfiable nor valid. Hence, the above formula is unsatisfiable.

Problem 4

For each pair of formulas below, determine by means of a truth table whether the two formulas are logically equivalent or not.

(a) $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$

(a) let $\phi = p \leftrightarrow q$ and $\psi = (p \wedge q) \vee (\neg p \wedge \neg q)$

p	q	$\neg p$	$\neg q$	$p \leftrightarrow q$	$p \wedge q$	$\neg p \wedge \neg q$	$(p \wedge q) \vee (\neg p \wedge \neg q)$
T	T	F	F	T	T	F	T
T	F	F	T	F	F	F	F
F	T	T	F	F	F	F	F
F	F	T	T	T	F	T	T

ϕ and ψ are logically equivalent. ($\phi \equiv \psi$)

(b) $p \leftrightarrow q$ and $(p \vee \neg q) \wedge (\neg p \vee q)$

(c) $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$

(d) $p \leftrightarrow (q \leftrightarrow r)$ and $(p \leftrightarrow q) \leftrightarrow r$

(b) $p \leftrightarrow q$ and $(p \vee \neg q) \wedge (\neg p \vee q)$

let $\phi = p \leftrightarrow q$, $\psi = (p \vee \neg q) \wedge (\neg p \vee q)$

p	q	$\neg p$	$\neg q$	$p \leftrightarrow q$	$p \vee \neg q$	$\neg p \vee q$	$(p \vee \neg q) \wedge (\neg p \vee q)$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	F	F
F	T	T	F	F	F	T	F
F	F	T	T	T	T	T	T

ϕ and ψ are logically equivalent. ($\phi \equiv \psi$)

(c) $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$

let $\phi = (p \wedge q) \rightarrow r$, $\psi = (p \rightarrow r) \wedge (q \rightarrow r)$

p	q	r	$p \wedge q$	$(p \wedge q) \rightarrow r$	$p \rightarrow r$	$q \rightarrow r$	$(p \rightarrow r) \wedge (q \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	T	T	T
T	F	F	F	T	F	T	F
F	T	T	F	T	T	T	T
F	T	F	F	T	T	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

ϕ and ψ are not logically equivalent. ($\phi \not\equiv \psi$)

(d) $p \leftrightarrow (q \leftrightarrow r)$ and $(p \leftrightarrow q) \leftrightarrow r$

let $\phi = p \leftrightarrow (q \leftrightarrow r)$, $\psi = (p \leftrightarrow q) \leftrightarrow r$

p	q	r	$q \leftrightarrow r$	$p \leftrightarrow (q \leftrightarrow r)$	$p \leftrightarrow q$	$(p \leftrightarrow q) \leftrightarrow r$
T	T	T	T	T	T	T
T	T	F	F	F	T	F
T	F	T	F	F	F	F
T	F	F	T	T	F	T
F	T	T	T	F	F	F
F	T	F	F	T	F	T
F	F	T	F	T	T	T
F	F	F	T	F	T	F

ϕ and ψ are logically equivalent. ($\phi \equiv \psi$)

Problem 5

(a) Consider the following argument:

$(v \wedge s) \rightarrow n$ ¹If Virginia supports independence, ²then so do the southern colonies. ³If Virginia and the southern colonies support independence, ⁴then the northern colonies will do the same. ⁵Therefore, ⁶Virginia's supporting of independence is sufficient for the northern and the southern colonies to both support independence.

Translate the sentences in this argument into formulas in propositional logic using the following atomic propositions:

- v = Virginia supports independence.
- s = The southern colonies support independence.
- n = The northern colonies support independence.

Then, check whether this argument is valid or not by means of a truth table.

$v \rightarrow s$
 $(v \wedge s) \rightarrow n$
 $\therefore v \rightarrow (s \wedge n)$

v	s	n	$v \wedge s$	Premise $(v \wedge s) \rightarrow n$	$s \wedge n$	Premise $v \rightarrow s$	Conclusion $v \rightarrow (s \wedge n)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	T	F
T	F	T	F	T	F	F	F
T	F	F	F	T	F	F	F
F	T	T	F	T	T	T	T
F	T	F	F	T	F	T	T
F	F	T	F	T	F	T	T
F	F	F	F	T	F	T	T

An argument is valid if and only if, whenever all the premises are true. $\therefore$ valid.

(b) Consider the following argument:

$$v \rightarrow s$$

$$(v \wedge s) \rightarrow n$$

¹If Virginia supports independence, then so do the southern colonies. ²If Virginia and the southern colonies support independence, then the northern colonies will do the same. ³Therefore, Virginia supports independence if and only if both the northern and the southern colonies also support independence.

$$v \leftrightarrow (s \wedge n)$$

Translate the sentences in this argument into formulas in propositional logic using the following atomic propositions:

v = Virginia supports independence.

s = The southern colonies support independence.

n = The northern colonies support independence.

Then, check whether this argument is valid or not by means of a truth table.

$$v \rightarrow s$$

$$(v \wedge s) \rightarrow n$$

$$\therefore v \leftrightarrow (s \wedge n)$$

				Premise $(v \wedge s) \rightarrow n$		Premise $v \rightarrow s$	Conclusion $v \leftrightarrow (s \wedge n)$
v	s	n	$v \wedge s$		$s \wedge n$		
T	T	T	T	T	T	T	T
T	T	F	T	F	F	T	F
T	F	T	F	T	F	F	F
T	F	F	F	T	F	F	F
F	T	T	F	T	T	T	F
F	T	F	F	T	F	T	T
F	F	T	F	T	F	T	T
F	F	F	F	T	F	T	T

$\therefore$ not valid

P is sufficient for Q $P \rightarrow Q$

P is necessary for Q $Q \rightarrow P$ or $\neg P \rightarrow \neg Q$

P is necessary and sufficient for Q $P \leftrightarrow Q$

$$((p \vee (\neg q)) \vee (r \wedge p)) \vee (q \wedge (\neg r))$$

p	q	r	$\neg q$	$\neg r$	$p \vee (\neg q)$	$r \wedge p$	$(p \vee (\neg q)) \vee (r \wedge p)$	$q \wedge (\neg r)$	$(p \vee (\neg q)) \vee (r \wedge p) \vee (q \wedge (\neg r))$
T	T	T	F	F	T	T	T	F	T
T	T	F	F	T	T	F	T	T	T
T	F	T	T	F	T	T	T	F	T
T	F	F	T	T	T	F	T	F	T
F	T	T	F	F	F	F	F	F	F
F	T	F	F	T	F	F	F	T	T
F	F	T	T	F	T	F	T	F	T
F	F	F	T	T	T	F	T	F	T

Since all are $\vee$, same precedence, equivalent but to keep it handy we grouped them into an easy way.

Also, logical OR ($\vee$) is associative and nothing's gonna change.

$$p \vee (\neg q) \vee (r \wedge p) \vee (q \wedge (\neg r))$$

p	q	r	$\neg q$	$\neg r$	$r \wedge p$	$q \wedge (\neg r)$	$p \vee (\neg q) \vee (r \wedge p) \vee (q \wedge (\neg r))$
T	T	T	F	F	T	F	T
T	T	F	F	T	F	T	T
T	F	T	T	F	T	F	T
T	F	F	T	T	F	F	T
F	T	T	F	F	F	F	F
F	T	F	F	T	F	T	T
F	F	T	T	F	F	F	T
F	F	F	T	T	F	F	T

$$E12 \quad \phi \wedge (\psi \wedge \chi) \equiv (\phi \wedge \psi) \wedge \chi$$

Associative Laws

$$E13 \quad \phi \vee (\psi \vee \chi) \equiv (\phi \vee \psi) \vee \chi$$